

Evaluating Soft Organ-Shaped Tangibles for Medical Virtual Reality

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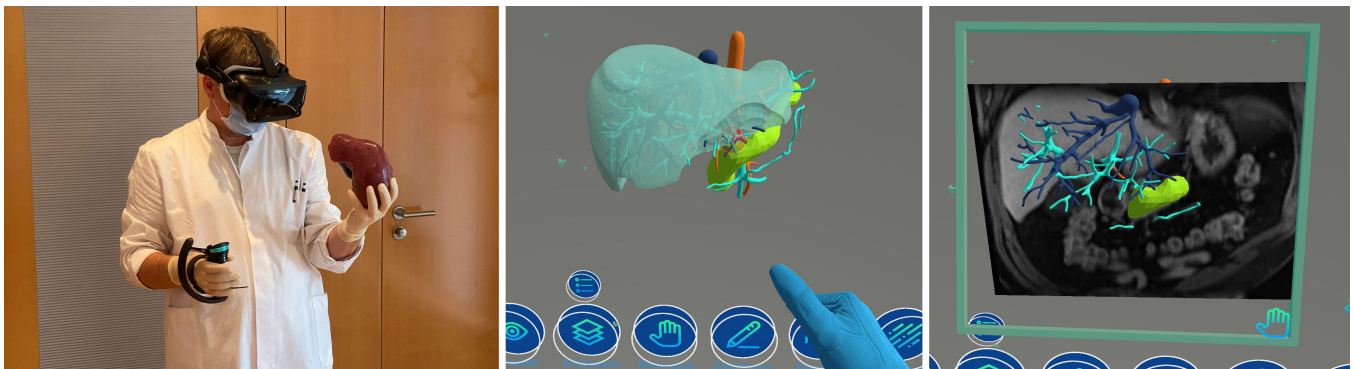


Figure 1: *Left image:* One visceral surgeon using the VR tool. *Centre & right image:* Two functions of the VR surgical planning tool: *centre* – the resection tool and *right* – the frame.

ABSTRACT

Connecting digital information with the physical is one of the essential ideas of tangible user interfaces. The design of the physical representation is important especially for specialised domains like surgery planning, because surgeons rely heavily on their tactile senses. Therefore, this research work investigates the effect of a soft and a hard 3D model as an interaction device for virtual reality surgical planning. A user study with 13 surgeons reveals a clear preference for the softer, more realistic material and a significantly

higher haptic user experience for the soft model compared to the hard one. These results advocate for stressing material aspects along with the interaction design in domains with an inherently high focus on tactile aspects.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; **Haptic devices**; **User studies**; • **Applied computing** → **Health care information systems**; • **Hardware** → **Haptic devices**.

KEYWORDS

virtual reality, surgery, surgical planning, natural interaction, 3D prints, model, liver, material

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1 INTRODUCTION

Giving physical form to digital information, providing a better integration of the physical and the computational worlds and allowing users to use their body skills when interacting with technology are central ideas of tangible and embodied interaction [11, 17, 22]. These ideas also matter for designing interaction in virtual reality (VR), which often lacks embodiment, haptic feedback, and rich materiality. Especially when it comes to expert applications in contexts, where the users have acquired extensive skills with their hands and where the profession relies on comprehensive experiences with specific work materials, computer systems should aim to incorporate these body skills and material experiences. Klemmer et al. [22], for example, highlight that especially in expert domains the users develop rich sensorimotor skills and tacit knowledge, which helps them in their daily work. Integrating these into human-computer interaction (HCI) is in line with the idea of reality-based interaction [18], in which interactions rely on pre-existing real world knowledge and skills. This has the potential to reduce the mental effort and to positively influence the gap described by the gulf of execution [16].

In VR-applications, we have seen approaches that integrate objects to enhance haptic feedback and manual interaction [5]. However, the experience of rich materiality in VR is still limited. While materiality in human-computer interaction has been addressed as an important topic on many levels [10, 47], only few works have systematically explored the experience of different physical materials for interaction yet, e.g. [19, 33]. Especially a systematical evaluation of micro level material aspects such as technical properties, addressed senses and usability, and macro level material aspects, such as material meanings in the application context and the evoked emotions, can enhance our understanding of improving the material selection for user interfaces [9].

In this work we investigate the material aspect for tangibles in VR in the specific domain of surgery planning. Surgical skills rely on sensory motor skills and tacit knowledge and therefore material aspects might play an important role also for surgical planning. Research works, which incorporate tangibles for surgery planning (e.g., [1, 4, 15, 36, 37, 39]) did not focus on the material aspect so far. They used standard hard 3D printing or other available materials like puppet heads. The materials generally used are far from feeling realistic regarding softness, and innovative specific controllers, such as the “Eye of Ra” device [4, 39], do not even match in form. Therefore, we want to evaluate the effect of softer, more realistic materials for surgery planning, in our case, liver surgery planning.

In contrast to the actual surgery, the planning currently still happens primarily in the digital domain and the minds of the surgeons. In this work we provide surgeons with a VR surgery planner, because VR can be especially helpful to view and interact with these 3D models as it provides the user a stereoscopic 3D view on the model and enables correct perception of depth. This enables a better spatial understanding of the anatomical structures and their relations [29]. In addition, VR can provide a sense of embodiment,

referring to the users sense of self-location and sense of agency [20] through intuitive controls to view the model just by head movement as well as the possibility to interact with objects through virtual hands. Examples for such systems are prototypes incorporating VR [2, 7, 8, 21, 36, 38, 43]. They present either 3D models or plain CT (computer tomography)/MRI (magnetic resonance imaging) images to the user and allow for various interaction methods, e.g. gestures, controller based or touch. There are also plenty of system supporting intraoperative image viewing and surgery support such as the ARTEMIS system [13, 46], but this work focuses on surgery planning.

In addition to the VR system, we incorporate 3D printed organ models as a tangible controller to plan the surgery. 3D prints are used to an increasing extent in medicine during the last decades [45]. A literature review by Martelli et al. [28] for the period of time between 2005 and 2015 showed that 71.5 % used it to produce anatomical models. The main advantage reported was better preoperative planning (48.7 %) and that it saved time in the operating room (32.9 %). One example of using 3D printed organ models is by Zein et al. [48]. They used 3D printed livers or parts of the liver for planning of a transplantation to make sure they discover potentially unsuitable patients based on the anatomical structure or that they are aware of any anomalies. Muguruza Blanco et al. [31] combined 3D printing and electroluminescence technologies, to provide a haptic and visual representation of relevant aspects for surgery planning. Fillat-Gomà et al. [12] also showed, that 3D planning and 3D printing in combination can decrease surgical time and surgical errors for upper extremity trauma cases. In these cases the 3D prints are mostly used without any kind of augmentation, but there exist several research works, that employed a combination of tangibles and display technologies such as 3D projection or VR to improve the interaction [1, 4, 15, 36, 37, 39]. Nevertheless, as mentioned before, most of them used only hard, standard 3D printing materials for their tangibles. As we expect an effect of softer, more realistic materials for tangibles in the context of surgical planning, we provide surgeons with a VR surgery planning tool and two 3D prints of identical size and weight but different material properties regarding softness. In a user study with the target audience of surgeons, we investigate the task load and user experience of the tangibles and VR tool. Thirteen (13) surgeons experienced both models in a counterbalanced study design and the majority (10) preferred the soft model over the hard one.

2 PROTOTYPE

The prototype used in this study has three “parts”. First a VR application, which incorporates planning tools for the second part – virtual organ models, specifically liver in this case. These virtual organ models are patient specific and can be controlled by the third part, tangibles which are general 3D prints of livers.

2.1 VR Prototype

The tool was developed using Unity 2019.4¹. It is focused on providing tools to plan a surgery on the patient specific 3D model of the diseased organ. It provides different tools and interactions for the planning informed by a user-centred approach. All interactions can

¹<https://unity.com/>

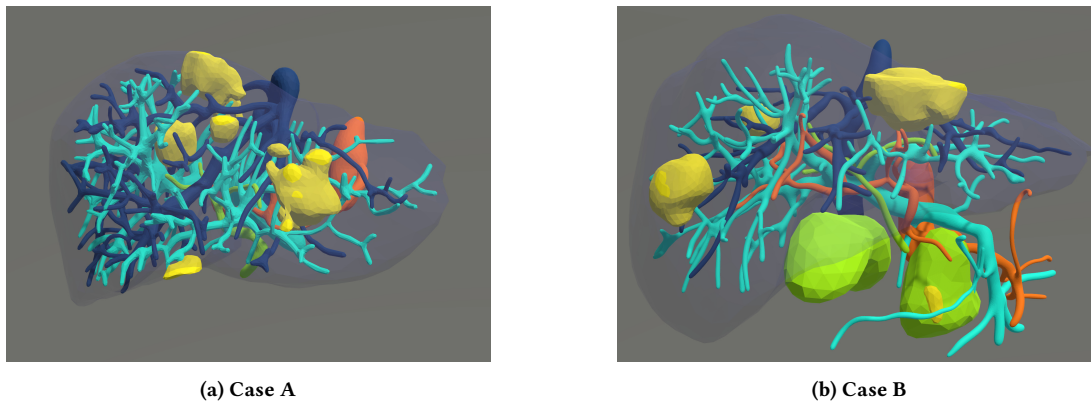


Figure 2: The virtual liver models

be performed with the index finger by pressing the trigger button on the controller and by opening and closing the hand (i.e. grip the controller). The application was designed around a one-button-style interaction and the concept of flat menus in order to keep it as simple as possible and make it easy to learn in a short time.

The tool provides the following functions: (1) display of the virtual organ, with and without the outer shell, (2) overlay of the MRI images using a frame, (3) a pen to mark aspects, (4) a measuring tool in form of a ruler, (5) a tissue remover in form of a sphere (see Figure 1 centre image), (6) a tool to place clips and (7) an option to associate the planning actions to more than one default step. Furthermore, the model can be scaled from 100 % to 300 % and the opacity of the outer shell can be adjusted from transparent to opaque. The last two functions are controlled by sliders placed under the menu.

The menu consisting of 3D buttons is present in mid-air to control the functions of the application. The interaction with the menu can be done from a distance through a pointer by using the trigger button of the VR controller. Additionally, the buttons can actually be pressed by moving the hand/finger close to the button and “press” it. The frame showing the radiological images can be grabbed and moved through the virtual model for browsing. Also the ruler for measuring can be grabbed and placed where needed. The functions “pen”, “clips” and “tissue remover” are attached to the hand upon activation – as a virtual pen, which paints when pressing the trigger, or as a sphere either removing tissue or a clip stapler, which can be differentiated by the sphere design containing a clip symbol or not. The percentage that the resected volume takes up of the whole model is calculated and presented to the user below the menu.

2.2 Virtual Models

The virtual models were created based on anonymised MRI images of patients. The images were annotated by (medical) radiological technologists using dedicated planning software and combined to a polygonal model. For further details of the process refer to [27, 35, 40]. The anatomical structures of the model, e.g., tumours, arteries, veins are coloured according to anatomy textbooks (see Fig. 2). For this study we chose three segmented liver models and integrated them and their MRI images manually into the system. One case

was just for training purposes and did not contain any lesions or tumours. The other two cases used in the study are depicted in Figure 3. Case A had several segmented tumour masses (yellow) in various liver segments and Case B three tumours which are located more on the outer layers of the liver. As the tangibles, i.e. 3D printed models, described in subsection 2.3 are not patient individual these tumour masses cannot be identified by touch. Besides the segmented tumours the liver models contain the main vessels or the venous (dark blue – vena cava, cyan – portal vein) and arterial (red/orange – arteria hepatica propria) system. Additionally, for Case B some vessels are extended to outside of the liver and the pancreas was part of the segmented model, but not incorporated in our study.

2.3 3D-printed Models

The focus of this research is on the experience of the tangible liver controller. We designed two liver tangibles in the size of roughly 75 % of a normal healthy liver. This size was preferred by experts in a study by Reinschluessel et al. [36]. The models are 19 cm (7.48 inch) and weight 820 g (roughly 29 oz). The solid liver controller is printed using laser sintering technique² using PA12 (see Figure 3a)³. This liver print was used to make a casting mould for the soft 3D model. For the soft one a smaller model (50 % of original size) was printed using the same laser sintering technique. A mount for a VIVE tracker was added and then it was placed in the casting mould. The mould was filled with a red coloured industrial gelatine mixture. After the gelatine has solidified it was coated with a latex mixture to stabilise and preserve it (see Figure 3b). Attached to both liver models is a 10 cm (3.9 inch) screw, each holding a VIVE tracker to connect it to the VR application. The virtual models were calibrated to best match with the tracked liver controller. As the tangibles were slightly smaller than their virtual counterparts (scaled to 100 % or more) we integrated a fine outline of the tangible in the VR to visualize where the tangible can be touched and grabbed.

²<https://xponentialworks.com/what-is-laser-sintering/>

³The printing material is white and because the participants are not seeing the models during the study, this model was not coloured differently.



(a) The printed liver with solid material



(b) The printed liver with soft material

Figure 3: The liver models (identical size and weight). Participants could not see the models. They only interacted with printed livers when wearing the VR-headset.

3 METHODS

To investigate the effects of more realistic softness for the liver model on the user experience, work load and interaction, we conducted a user study.

3.1 Participants

Thirteen (13) participants (3 female, 10 male) participated in our study. The participants were recruited at a local university hospital using purposive sampling. Our participants were all assistant doctors ($n = 5$), consulting physicians ($n = 2$) and senior/head physicians ($n = 6$). Due to privacy reasons the age and experience was asked as categorical variables. The participants reported being in the following age ranges: 26-30 years ($n = 2$), 31-35 years ($n = 2$), 36-40 years ($n = 2$), 41-45 years ($n = 3$), 46-50 years ($n = 3$), and 56-60 years ($n = 1$). Their experience in the surgical field is the following: 0-5 years ($n = 3$), 6-10 years ($n = 3$), 11-15 years ($n = 4$), 16-20 years ($n = 2$), and 21-25 years ($n = 1$). The majority of the participants had little previous experience with VR: eight (8) of them used VR up to 5 times before and one (1) did not have any previous VR experience. Three (3) participants used VR up to 15 times before and one (1) even used VR more than 15 times before. For the previous experience with 3D models (either on a 2D screen or in VR) four (4) had no experience, four (4) used up to 10 times and four (4) used it up to 20 times before. One participant has used 3D models more than 20 times. It is noteworthy that two participants reported not being able to see three dimensional.

3.2 Setup

The VR setup consisted of a VALVE Index⁴ and a computer with Intel i7 processor and a Nvidia 2080 TI graphics card. Room-scaled tracking was used with an approximate area of 3 x 3 meter. The application was controlled with one VALVE Index controller, which

was held in the dominant hand of the participant and the tangible in the other hand.

3.3 Procedure

The study received approval from the ethical board of the university hospital. After signing the applicable consent forms, the participants were informed about the study process and got an introduction into the VR hardware. For hygiene reasons all participants and study coordinators wore surgical gloves, which also ensures a realistic feeling of the haptic properties as experienced during surgery itself. The study coordinators assisted the participants in fitting the VR headset and the VALVE controller. When the hardware fit to the satisfaction of the participants, the study coordinators started the introduction to the VR tool presented in subsection 2.1 using an exemplary case.

Following this, the participants started the first round of two. The participants got one of the two liver controllers into their free hand – usually the non-dominant hand – to interact with the virtual liver together with the VALVE controller. The conditions were pseudo-randomised using a Latin-square scheme to balance biasing effects of the virtual and physical model order and combination. To guide the participants through the study, they were given tasks which were identical for both conditions. The first task was to identify potentially variants of the vessel system. Next, they had to identify the segments of the tumours and measure the diameter of them. Further, they were asked to show the location of one tumour in the MRI images. The study instructors also asked the participants to remove the tissue around the tumour and put clips on the open vessels. Lastly, the surgeons had to assess the general state of the participants, if all tumours were removed. Even though all participants were visceral surgeons, they were not necessarily experts on livers. Therefore, the answers to these tasks were neither reported nor analysed as they depended on the training and the focus of this work is on the interaction. Yet, these tasks were used to motivate the exploration of the model (see Figure 1 left image). After

⁴<https://www.valvesoftware.com/en/index>

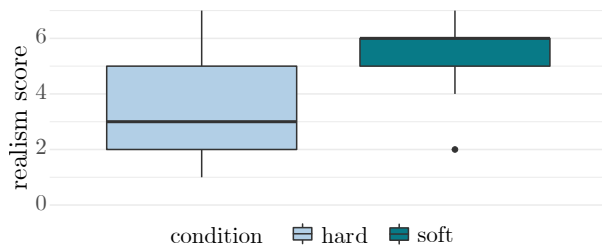


Figure 4: Realism scores for the two models – 7 means highly realistic, 1 means not realistic at all

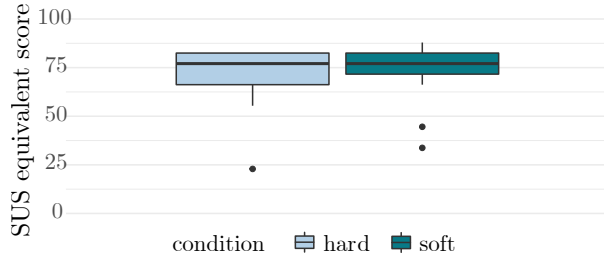


Figure 5: UMUX scores scaled to the 0 to 100 scale of the SUS

finishing the tasks, each participant was asked about their intuitive judgement of the physical model they used. Each of the two rounds was concluded with a set of questionnaires: the UMUX lite [25], the NASA Task Load Index (TLX) [14] and the UEQ+ [42], which consisted of the normal UEQ [41] and four additional scales about haptics, personalisation, usefulness and intuitive use. Additionally, we asked about the perceived realism of the model on a Likert scale from 1 (not realistic at all) to 7 (highly realistic). When the participants completed both rounds, they received a demographic questionnaire and were interviewed about their experience using a semi-structured interview. The procedure took between 20 and 50 minutes depending on the previous experience of the participants. The participants received no compensation for their participation. The interaction during the study was video recorded and the interviews were voice recorded for future analysis.

3.4 Analysis

The questionnaires were analysed using R [34]. We analysed the NASA TLX and the self-design scale for realism using a paired t-test as the data were normal distributed according to the Shapiro-Wilk test. The UEQ(+) scales and the UMUX Lite were analysed using the non-parametric Wilcoxon signed-rank test.

4 RESULTS

Figure 4 shows the results for the question about the realism of the two models tested. The soft model achieved an average realism score of 5.54 (SD = 1.34) and the hard model had an average realism score of 3.62 (SD = 1.94), while 7 is equal to very realistic and 1 to not realistic at all. The t-test showed a significant difference $t(13) = 4.2696$ and $p = .0009$.

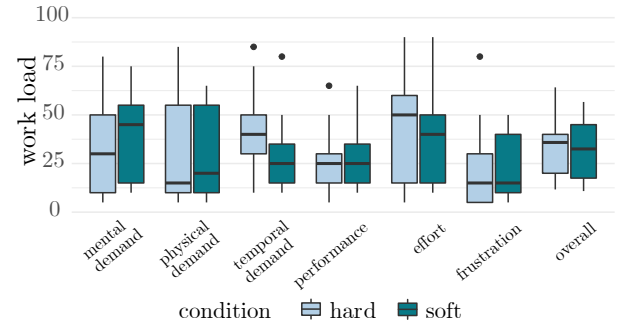


Figure 6: NASA TLX scores for each model

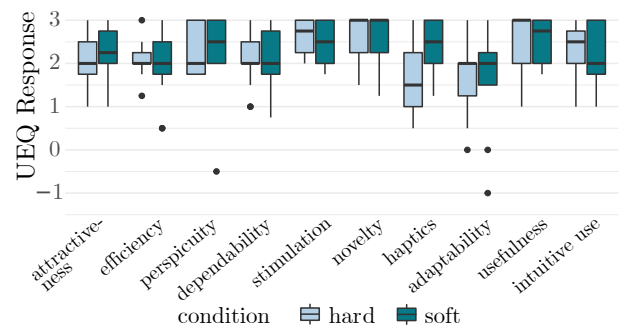


Figure 7: UEQ(+) scores for the models – scale from -3 to 3

The results for the NASA TLX questionnaire are depicted in Figure 6. The statistical analysis revealed no significant differences between the models, although the temporal demand had a p -value of .069.

Figure 7 shows the answer distribution for the UEQ(+) scales. The difference regarding the haptic scale is significant ($p = .0075$). On a scale from -3 to 3 the soft model received on average a 2.3654 (median = 2.5, SD = .6260) while the hard model had a mean of 1.6346 (median = 1.5673, SD = .7634). The other scales did not exhibit any significant differences.

The scores from the UMUX Lite [24] converted to SUS scores [6, 26] are presented in Figure 5. Both models achieved similar results. For the soft one a mean of 73.3167 (median = 77.0667, SD = 16.7515) was reported, while the hard one had a mean of 69.9833 (median = 77.0667, SD = 16.6458).

When asked about their preferred model, ten (10) participants preferred the soft one, two (2) the hard one and one (1) could not decide as both “did not feel natural”. Overall the participants liked the soft model, because it felt more realistic and it is easier to grasp because of the material, which is more non-skid than the solid print. Some participants mentioned that the models felt different in size and weight. Overall, they all like using a haptic model to interact with the virtual model, because it is intuitive to interact with. Nevertheless, some mentioned that the model gets a little heavy over time and would have preferred a lighter model. When asked about other surgery scenarios of 3D printed models, they mentioned spleen, rectum, and lung surgery. In addition, the

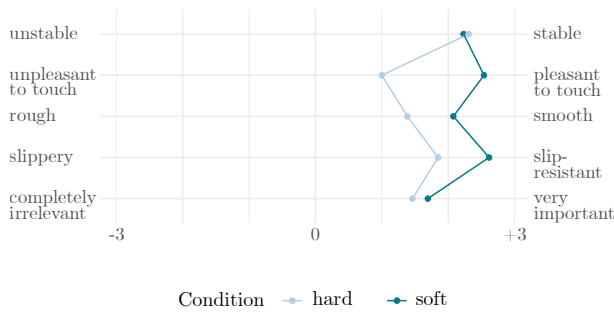


Figure 8: Visualisation of the means for the UEQ+ question regarding haptics.

participants mentioned that the tool and tangibles could be used in education for anatomy learning.

5 DISCUSSION

The results show that the surgeons had a clear preference for interacting with the softer, more realistic feeling tangible. They also rated the soft liver model as significantly more realistic even though the softness and texture did not match a real liver perfectly. These results are in line with findings from Muender et al. [30], who found that users prefer to interact with more realistic tangibles. Their results also show that the haptic fidelity has an influence on the performance and accuracy of the interaction. In this study both tangibles were similar in shape, size and weight and we did not observe an improvement in measurements like performance, task load (including performance) or usability. Thereby we add evidence to the previously published results, that given a certain realism like in Muender et al. [30], the general usability and performance does not necessarily improve and usage without measurable restrictions is possible. Nevertheless, the haptics question of the UEQ+ shows a significant difference indicating higher haptic qualities for the soft liver models. This shows that the effort to create a more realistic and relatable haptic experience was appreciated by the experts and its effect should be investigated in more detail.

In the interviews, the surgeons also reported that the soft tangible was easier to grasp due to its soft material. This is also supported by the question regarding the slip-resistance of the UEQ+ that also had significantly higher values for this single question ($p = .0247$, see Figure 8). This shows that the design choice for a particular material including its properties not only affects user experience aspects such as perceived realism, but also practical aspects such as the handling. While the results in this study recommend to decide for the more realistic material, a previous study on the aspect of size [36] showed that regarding this detail the more realistic choice is not necessarily better. For the design of tangibles a holistic approach should be taken, as various aspects like shape, size or material properties interact creating the final user experience and usability.

The overall weight of both versions of tangibles was reported to be a bit heavy for comfortable interaction by some participants. The weight was mainly determined by the process to create the soft tangible by casting gelatin and the tangibles' size, which was chosen according to results from a previous study [36]. While the

hard tangible could have been much lighter, we adjusted the weight to match the soft tangible for comparability. The manufacturing of the soft tangible with gelatin was chosen to achieve the most realistic haptic properties resulting in a uniform softness for the whole tangible. Other works have introduced interfaces that allow for dynamic changes of softness properties, which could be an interesting approach to create a patient specific tangible that adapts to the patient's circumstances. As the softness of healthy liver tissue is mostly the same for humans, the display of diseased tissue that has a different softness would be of particular interest. This would require to have varying softness in different parts of the organ rather than a uniform change of softness. However, current research mainly targets the whole surface [3, 32] or only a small area on the fingertip [23] while it would be important to have a feedback to all fingers and the palm when holding the organ tangible. A possible approach to achieve dynamic changes in softness is presented by Tsimmeris et al. [44]. Enabling surgeons to palpate a liver tangible with realistic representation of different tissue softness could make the virtual preparation for surgery even more useful.

6 CONCLUSION & FUTURE WORK

In this work we investigate the effects of soft, more realistic material for tangibles in the context of surgery planning. In a user study with 13 surgeons we compare two types of liver shaped tangibles with soft and solid material to interact with a VR surgical planning software. The results show a clear preference for the softer material, which was perceived as more realistic by the surgeons while the usability and task load did not differ between the conditions. As next steps, we plan to increase our sample with additional surgeons and analyse the interaction with the model in detail using video recordings. In the future, we will also plan to investigate different thicknesses of the gelatin layer to improve the weight of the tangibles and further investigate how the tangibles can be improved to provide benefits for a more intuitive and productive surgical planning process.

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